MODULE 4 — 7-Mark Questions with Answers

Exam-oriented answers | No diagrams | Important definitions, explanations, formulas, examples, and comparisons

1. Explain SNMP architecture and working, including manager, agent, operations, SMI, and MIB.

SNMP (Simple Network Management Protocol) is an application-layer protocol used to monitor, manage, and control network devices such as routers, switches, and servers.

Architecture and components

1. SNMP Manager: A centralized management system that sends requests to managed devices, receives responses, monitors performance, and detects faults.
2. SNMP Agent: Software running on a managed device that collects device information, processes management requests, and sends notifications to the manager.
3. Managed Device: A router, switch, server, or other network device being monitored.
4. MIB (Management Information Base): Organizes information about manageable objects, such as interface status, packet counts, and device uptime.
5. SMI (Structure of Management Information): Defines the rules, data types, naming conventions, and structure used to describe MIB objects.

Working of SNMP

1. The manager sends a request to the agent to retrieve or modify information.
2. The agent accesses the relevant managed objects and obtains their values.
3. The agent sends a response containing the requested information or operation status.
4. The manager analyzes the information and can take corrective action.
5. The agent can also send unsolicited notifications when important events occur.

Three basic management operations

- Get: Retrieves the value of a managed object.
- Set: Changes the value of a supported writable managed object.
- Trap: Allows the agent to notify the manager about an event, such as a link failure, without waiting for a request.

Other SNMP operations include GetNext, GetBulk, Response, and Inform.

Relationship between SNMP, SMI, and MIB

- SNMP provides the communication protocol.
- SMI defines how management information is named and represented.
- MIB organizes the definitions of manageable objects.

Example: To check a router's interface status, the manager uses SNMP to request the value of an object defined in the MIB according to SMI rules.

Conclusion: SNMP provides centralized network monitoring and management, while SMI and MIB provide the structure and information required for management.

2. What is ASN.1? Explain its need, built-in data types, new types, subtypes, SEQUENCE, and SEQUENCE OF.

ASN.1 (Abstract Syntax Notation One) is a standardized notation used to define data structures and data types exchanged between different computer systems.

Need for ASN.1

- Provides a standard method of describing data.
- Supports communication between systems using different hardware and software.
- Defines the structure and types of exchanged information.
- Helps protocols represent complex data consistently.

Built-in data types

Common ASN.1 types include:

- `INTEGER` — represents integer values.
- `BOOLEAN` — represents TRUE or FALSE.
- `REAL` — represents real numbers.
- `NULL` — represents the absence of a value.
- `BIT STRING` — represents a sequence of bits.
- `OCTET STRING` — represents a sequence of bytes.
- `IA5String` — represents characters from the IA5 character set.
- `ENUMERATED` — represents one value from a defined list.

Creating new types

A new type is defined using the assignment operator `::=`.

Example:

text
```
Age ::= INTEGER
```

This creates a new type called `Age` based on `INTEGER`.

Creating subtypes

A subtype restricts the permitted values or size of an existing type.

Example:

text
```
Age ::= INTEGER (0..150)
```

This restricts Age values to integers between 0 and 150.

Another example:

text
```
Name ::= IA5String (SIZE (1..30))
```

This allows strings containing 1 to 30 characters.

SEQUENCE

`SEQUENCE` combines fields of different types into a structured record.

Example:

text
```
Student ::= SEQUENCE {
    id INTEGER,
    age Age,
    name Name
}
```

It defines a student record containing an ID, age, and name.

SEQUENCE OF

`SEQUENCE OF` represents an ordered collection of elements of the same type.

Example:

text
```
StudentList ::= SEQUENCE OF Student
```

This defines a list containing multiple Student records.

Conclusion: ASN.1 provides a standardized way to describe simple values, restricted types, structured records, and collections for reliable data exchange.

3. Discuss data and signals at the physical layer. Explain analog and digital signals, sine-wave parameters, bandwidth, and Fourier analysis.

The physical layer is the lowest layer of the OSI model. It transmits raw bits through a communication medium using electrical, optical, or electromagnetic signals.

Analog and digital signals

Analog signal: A continuous signal that varies smoothly with time and can take infinitely many values within a range.

Examples: Human voice and temperature variations.

Digital signal: A signal represented using discrete levels, commonly corresponding to binary 0 and 1.

Examples: Binary data transmitted between computers.

Three parameters of a sine wave

A sine wave can be represented as:

s(t)=Asin⁡(2πft+ϕ)s(t)=A\sin(2\pi ft+\phi)

1. Amplitude (A): Maximum magnitude of the signal measured from its reference level. It indicates signal strength.
2. Frequency (f): Number of cycles per second, measured in hertz (Hz).
3. Phase (ϕ\phi): Indicates the relative position of the waveform with respect to a reference.

The time period and frequency are related by:

T=1fT=\frac{1}{f}

Bandwidth

Bandwidth is the difference between the highest and lowest frequencies in a signal.

B=fmax⁡−fmin⁡B=f_{\max}-f_{\min}

For example, if a signal occupies frequencies from 1000 Hz to 5000 Hz:

B=5000−1000=4000 HzB=5000-1000=4000\text{ Hz}

Therefore, its bandwidth is 4 kHz.

Composite signals and Fourier analysis

A composite signal consists of two or more sine waves of different frequencies, amplitudes, or phases.

Fourier analysis states that a suitable periodic signal can be represented as a sum of sinusoidal components. More general signals can be analyzed using Fourier transforms.

For example, a square wave can be represented as a sum of a fundamental frequency and odd harmonics with decreasing amplitudes.

The frequency components determine the bandwidth required to transmit the signal accurately.

Conclusion: The physical layer transmits signals through a medium. Understanding signal parameters, bandwidth, and Fourier analysis helps determine the capacity and performance of communication systems.

4. Explain attenuation, distortion, and noise. State the Nyquist and Shannon formulas and calculate an example for each.

Transmission impairment occurs when the received signal differs from the transmitted signal.

1. Attenuation

Attenuation is the reduction in signal strength as it travels through a medium.

- It increases with distance and depends on the transmission medium.
- It may be compensated for using suitable amplifiers or repeaters.
- Example: A weak signal received after travelling through a long cable.

Power gain or loss in decibels is:

dB=10log⁡10(PrPt)\text{dB}=10\log_{10}\left(\frac{P_r}{P_t}\right)

Here, PrP_r is received power and PtP_t is transmitted power.

2. Distortion

Distortion occurs when the shape of a signal changes during transmission.

- Different frequency components may experience different attenuation or propagation delays.
- The received waveform may become different from the original waveform.
- Example: Digital pulses spreading into adjacent bit intervals.

3. Noise

Noise is unwanted energy added to a transmitted signal.

Common types include thermal noise, induced noise, crosstalk, and impulse noise.

Example: Electrical interference causing unwanted sounds during a telephone call.

Nyquist bit rate formula

For an ideal noiseless, band-limited channel:

Rmax=2Blog2L\boxed{R_{\max}=2B\log_2 L}

Where:

- BB = bandwidth in Hz.
- LL = number of distinct signal levels.
- Rmax R_{\max} = maximum theoretical bit rate in bps.

Example: Let bandwidth be 3000 Hz and the number of signal levels be 4.

Rmax=2(3000)log24R_{\max}=2(3000)\log_2 4
=6000×2=12,000 bps=6000\times2=\boxed{12,000\text{ bps}}

Shannon capacity formula

For a channel affected by additive white Gaussian noise:

C=Blog2(1+SN)\boxed{C=B\log_2\left(1+\frac{S}{N}\right)}

Where S/NS/N is the signal-to-noise power ratio in linear form.

Example: Bandwidth = 3000 Hz and SNR = 30 dB.

First convert SNR to linear form:

SN=1030/10=1000\frac{S}{N}=10^{30/10}=1000

Then:

C=3000log2(1+1000)C=3000\log_2(1+1000)
C≈29,902 bps\boxed{C\approx29,902\text{ bps}}

Difference between Nyquist and Shannon

- Nyquist assumes a noiseless channel and considers bandwidth and signal levels.
- Shannon considers noise and establishes the theoretical channel capacity based on bandwidth and SNR.
- Nyquist is useful for understanding signaling limits, whereas Shannon determines the fundamental capacity limit for the assumed noise model.

Conclusion: Transmission impairment affects signal quality, while Nyquist and Shannon formulas help determine theoretical data transmission limits.

5. Describe PCM, including sampling, quantization, encoding, bit-rate derivation, and comparison with Delta Modulation.

Pulse Code Modulation (PCM) is a technique used to convert an analog signal into digital form.

1. Sampling

Sampling measures the amplitude of an analog signal at regular time intervals.

According to the Nyquist sampling theorem, the sampling rate must be at least twice the highest frequency of a band-limited signal for ideal reconstruction.

fs≥2fmaxf_s\geq2f_{\max}

For a voice signal with a maximum frequency of 4000 Hz:

fs=2(4000)=8000 samples/sf_s=2(4000)=8000\text{ samples/s}

2. Quantization

Quantization assigns each sample to the nearest available amplitude level.

If the number of quantization levels is LL, the number of bits per sample is:

n=log2Ln=\log_2 L

For a uniform quantizer with step size Δ\Delta, the maximum absolute quantization error when rounding to the nearest level is:

Emax=Δ2\boxed{E_{\max}=\frac{\Delta}{2}}

Quantization introduces a small difference between the original sample and its represented value.

3. Encoding

Each quantized level is assigned a binary code word. If 8 bits are used per sample, the number of possible code levels is:

L=28=256L=2^8=256

PCM bit-rate calculation

For standard narrowband voice PCM:

- Sampling rate = 8000 samples/s.
- Bits per sample = 8.

The bit rate is:

Rb=fs×nR_b=f_s\times n

Rb=8000×8R_b=8000\times8
Rb=64,000 bps=64 kbps\boxed{R_b=64,000\text{ bps}=64\text{ kbps}}

Thus, the PCM bit rate is 64 kbps per voice channel.

PCM versus Delta Modulation (DM)

PCM	Delta Modulation
Encodes the quantized amplitude of each sample.	Encodes whether the signal has increased or decreased relative to the previous estimate.
Usually uses multiple bits per sample.	Typically uses one bit per sample.
Requires a relatively higher bit rate.	Can require a lower bit rate at comparable sampling rates.
Quantization error occurs.	Slope overload distortion and granular noise may occur.
More accurate representation with suitable quantization.	Simpler implementation but may require oversampling.

Conclusion: PCM provides accurate digital representation through sampling, quantization, and encoding. Delta Modulation uses simpler one-bit decisions but is more susceptible to slope overload and granular noise.

6. Explain line coding, block coding, and scrambling. Describe polar, bipolar, multilevel, 4B/5B, 8B/10B, B8ZS, and HDB3.

Digital-to-digital conversion transforms digital data into signal patterns suitable for transmission.

A. Line coding

Line coding maps bits directly into digital signal levels or transitions.

1. Polar coding

Polar coding uses positive and negative voltage levels.

- NRZ-L: The voltage level represents the bit.
- NRZ-I: A transition at the start of the bit interval represents a 1 under the usual convention.
- RZ: The signal returns to zero during each bit interval.
- Manchester: Every bit has a transition in the middle, helping synchronization.
- Differential Manchester: A mid-bit transition provides clocking, while the presence or absence of a boundary transition represents data.

2. Bipolar coding

Bipolar coding uses three levels: positive, zero, and negative.

- AMI: Binary 0 is represented by zero voltage; successive binary 1s alternate between positive and negative voltages.
- This reduces the DC component and can help identify some transmission errors.

3. Multilevel coding

Multilevel schemes use more than two signal levels or encode groups of bits into signal elements.

Examples include 2B1Q, which maps two bits into one of four signal levels, and 8B6T, which maps eight bits into six ternary signal elements.

These techniques can reduce the number of signal elements required for a given bit rate, but receiver complexity may increase.

B. Block coding

Block coding maps groups of mm data bits to groups of nn code bits, where n>mn>m.

Its purposes include improving synchronization, controlling run lengths, and providing additional patterns that may support error detection.

1. 4B/5B

Every 4 data bits are mapped to a 5-bit code group.

Example:

```
0000 → 11110
```

The code set is selected to limit long runs of zeros, helping provide transitions when the encoded data is transmitted using NRZ-I.

Efficiency:

η=45×100=80%\eta=\frac{4}{5}\times100=\boxed{80\%}

2. 8B/10B

Every 8 data bits are mapped to a 10-bit code group.

- Provides controlled run lengths and sufficient transitions for clock recovery.
- Helps maintain DC balance by selecting suitable code groups.
- Includes special control characters.
- Commonly used in high-speed serial communication systems.

Its data-code efficiency is:

$\eta = \frac{8}{10} \times 100 = \boxed{80\%}$

C. Scrambling

Scrambling replaces certain bit patterns with predefined patterns to prevent long sequences of zeros in a line-coded signal. Unlike block coding, it generally preserves the bit rate and is reversed at the receiver.

1. B8ZS

B8ZS means Bipolar with 8-Zero Substitution.

- Used with bipolar AMI coding.
- Replaces eight consecutive zeros with a special pattern containing bipolar violations.
- The receiver recognizes the pattern and restores the original eight zeros.
- Helps maintain synchronization without adding extra bits.

2. HDB3

HDB3 means High-Density Bipolar 3-Zero.

- Replaces each run of four consecutive zeros with a special pattern.
- Uses bipolar violations and balancing rules to preserve synchronization and limit DC buildup.
- Commonly used in digital telephone transmission systems.

Conclusion: Line coding represents bits as signal levels, block coding maps data groups to controlled code groups, and scrambling replaces problematic bit patterns to maintain synchronization.

7. Explain ASK, FSK, PSK, and QAM, including bandwidth formulas, constellation diagrams, and noise performance.

Digital-to-analog conversion represents digital data by modifying the properties of an analog carrier signal.

1. Amplitude Shift Keying (ASK)

In ASK, the amplitude of the carrier changes according to the digital input.

- Binary 1 may be represented by a carrier with a particular amplitude.
- Binary 0 may be represented by a lower amplitude or no carrier.
- It is simple but sensitive to amplitude noise.

For binary ASK with rectangular pulse shaping, an approximate first-null bandwidth is:

$B \approx 2R_b$

The exact occupied bandwidth depends on pulse shaping.

2. Frequency Shift Keying (FSK)

In FSK, the carrier frequency changes according to the digital data while the amplitude remains constant.

- Binary 1 uses one frequency.
- Binary 0 uses another frequency.
- It is generally more resistant to amplitude variations than ASK but often requires more bandwidth.

For binary FSK, a commonly used approximate bandwidth expression is:

B≈2Δf+RbB\approx2\Delta f+R_b

Here, Δf\Delta f is the maximum frequency deviation from the center frequency. The precise bandwidth depends on the modulation index and pulse shaping.

3. Phase Shift Keying (PSK)

In PSK, the carrier phase changes according to the input data.

For BPSK, the two symbols differ in phase by 180∘180^\circ.

For ideal BPSK with rectangular pulses, the approximate first-null bandwidth is:

B≈2RbB\approx2R_b

PSK generally offers better power efficiency than ASK in amplitude-noise-limited channels.

4. Quadrature Amplitude Modulation (QAM)

QAM varies both amplitude and phase by combining in-phase and quadrature carrier components.

The number of bits per symbol is:

k=log⁡2Mk=\log_2 M

For 16-QAM:

k=log⁡216=4 bits/symbolk=\log_2 16=4\text{ bits/symbol}

For ideal linear QAM with raised-cosine pulse shaping, the approximate passband bandwidth is:

B≈(1+α)RsB\approx(1+\alpha)R_s

where RsR_s is the symbol rate and α\alpha is the roll-off factor.

Constellation diagram

A constellation diagram plots possible symbols using:

- X-axis: In-phase component (I).
- Y-axis: Quadrature component (Q).

Each point represents a symbol. BPSK has two constellation points, while 16-QAM has 16 possible points.

Why PSK and QAM are preferred over ASK in noisy channels

- ASK depends directly on carrier amplitude, so amplitude noise can cause decision errors.
- PSK uses phase differences and is less affected by pure amplitude variations.
- QAM transmits multiple bits per symbol and can provide high spectral efficiency when the signal-to-noise ratio is adequate.
- Higher-order QAM requires a higher SNR because its constellation points are closer together.

Conclusion: ASK is simple, FSK changes frequency, PSK changes phase, and QAM changes both amplitude and phase. PSK and QAM often offer better power or spectral efficiency than ASK, depending on channel conditions.

8. Explain AM, FM, and PM, their bandwidth formulas, and why FM is more resistant to noise than AM.

Analog-to-analog conversion changes a carrier signal according to a continuous information signal.

1. Amplitude Modulation (AM)

In AM, the carrier amplitude varies in proportion to the message signal, while the carrier frequency remains constant.

The conventional AM signal contains a carrier and two sidebands.

For a single-tone message:

BAM=2fm\boxed{B_{AM}=2f_m}

where fmf_m is the highest modulating frequency.

Example: Conventional AM radio broadcasting.

2. Frequency Modulation (FM)

In FM, the instantaneous frequency of the carrier varies according to the message signal, while the carrier amplitude remains constant.

Using Carson's rule, the approximate bandwidth for single-tone FM is:

BFM≈2(Δf+fm)\boxed{B_{FM}\approx2(\Delta f+f_m)}

where Δf\Delta f is the peak frequency deviation and fmf_m is the highest modulating frequency.

Example: FM radio broadcasting.

3. Phase Modulation (PM)

In PM, the instantaneous phase of the carrier varies directly according to the message signal. The carrier amplitude remains constant.

For a sinusoidal message, the approximate bandwidth using Carson's rule is:

BPM≈2(Δf+fm)\boxed{B_{PM}\approx2(\Delta f+f_m)}

Here, Δf\Delta f is the peak frequency deviation produced by the phase modulation for that message frequency. Unlike FM, the deviation depends on the message frequency as well as the phase sensitivity.

Comparison table

Property	AM	FM	PM
Changed property	Amplitude	Instantaneous frequency	Instantaneous phase
Carrier amplitude	Varies	Constant	Constant
Carrier frequency	Constant in the unmodulated carrier reference	Varies instantaneously	May vary instantaneously as phase changes
Bandwidth	2fm2f_m for conventional single-tone AM	Approximately 2(Δf+fm)2(\Delta f+f_m)	Approximately 2(Δf+fm)2(\Delta f+f_m) using Carson's rule
Noise performance	More sensitive to amplitude noise	Better rejection of amplitude noise	Better rejection of amplitude noise

Why FM is more resistant to noise than AM

1. FM information is carried by frequency variations rather than amplitude variations.
2. Amplitude noise can be reduced using limiter circuits before demodulation.
3. FM systems can use pre-emphasis and de-emphasis to improve noise performance.
4. FM generally provides better noise performance at the cost of greater bandwidth in many applications.

Conclusion: AM varies amplitude, FM varies instantaneous frequency, and PM varies instantaneous phase. FM is more resistant to amplitude noise because its information is carried primarily in frequency variations.

9. Explain bandwidth utilization techniques: FDM, WDM, TDM, FHSS, and DSSS. Distinguish multiplexing from spread spectrum.

Bandwidth utilization techniques help communication systems use available transmission resources efficiently.

A. Multiplexing techniques

Multiplexing combines multiple signals for transmission over a shared physical medium.

1. Frequency Division Multiplexing (FDM)

FDM divides the available channel bandwidth into separate frequency bands, assigning each signal a different band.

- Signals can be transmitted simultaneously.
- Guard bands may be used to reduce interference.
- Example: Radio and cable television broadcasting.

2. Wavelength Division Multiplexing (WDM)

WDM is used in fiber-optic communication. Multiple optical signals with different wavelengths are combined and transmitted through the same fiber.

- Each wavelength carries a separate data channel.
- It greatly increases the capacity of optical fiber.
- Example: Long-distance optical networks.

3. Time Division Multiplexing (TDM)

TDM allows multiple signals to share a channel by assigning each signal a time slot.

Synchronous TDM:

- Each source receives a fixed time slot in every frame.
- Slots may be wasted when a source has no data.
- Timing is predictable.

Statistical TDM:

- Time slots are allocated dynamically to active sources.
- Channel utilization is improved for bursty traffic.
- Addressing or equivalent source identification and buffering may be required.

B. Spread spectrum techniques

Spread spectrum deliberately spreads a signal over a bandwidth wider than the minimum required for its data.

1. Frequency Hopping Spread Spectrum (FHSS)

- The carrier hops between frequencies according to a pseudorandom sequence.
- The receiver follows the same sequence.
- It can avoid some narrowband interference by changing frequencies.
- Example: Bluetooth Classic.

2. Direct Sequence Spread Spectrum (DSSS)

- Data is combined with a high-rate spreading code.
- The transmitted signal occupies a wider bandwidth.
- The receiver uses the same code to despread the signal.
- Example: DSSS-based IEEE 802.11b Wi-Fi modes.

Multiplexing versus spread spectrum

Multiplexing	Spread spectrum
Combines multiple data streams on a shared medium.	Spreads a signal across a wider bandwidth.
Main goal is efficient sharing of channel capacity.	Main goals include interference resistance, robustness, and processing gain.
Separates signals by frequency, wavelength, or time.	Uses frequency hopping or spreading codes.
Examples: FDM, WDM, TDM.	Examples: FHSS, DSSS.

Conclusion: Multiplexing improves sharing of a communication medium, whereas spread spectrum expands the signal's bandwidth to improve robustness and interference performance.

10. Discuss guided and unguided transmission media with construction, working, advantages, applications, and comparison.

Transmission media carry data from a sender to a receiver. They are classified as guided and unguided media.

A. Guided transmission media

Guided media use a physical path to carry signals.

1. Twisted-pair cable

Construction: Consists of two insulated copper wires twisted together. It may be unshielded (UTP) or shielded (STP).

Working: Electrical signals travel through the copper wires. Twisting helps reduce electromagnetic interference and crosstalk.

Advantages:

- Low cost.
- Easy to install and maintain.
- Widely available.

Applications: Ethernet LANs, telephone networks, and office communication systems.

2. Coaxial cable

Construction: Contains a central copper conductor, insulating dielectric, metallic shield, and outer protective jacket.

Working: Electrical signals travel through the central conductor, while the metallic shield helps contain the signal and reduce external interference.

Advantages:

- Good shielding and noise immunity.
- Suitable for higher frequencies than many basic copper links.
- Durable and reliable.

Applications: Cable television, broadband internet access, and radio-frequency connections.

3. Fiber-optic cable

Construction: Consists of a glass or plastic core, cladding, and protective outer layers.

Working: Data is transmitted as pulses of light through the core. The cladding supports light confinement through total internal reflection under suitable conditions.

Advantages:

- Very high bandwidth.
- Low attenuation over long distances.
- Immune to electromagnetic interference.
- Provides electrical isolation.

Applications: Internet backbones, long-distance telecommunications, and data centers.

B. Unguided transmission media

Unguided media transmit electromagnetic waves through air or space without a physical cable.

1. Radio waves

Working: Radio-frequency electromagnetic waves propagate through space, often using antennas that provide broad coverage.

Advantages: Can cover large areas, and suitable frequencies can penetrate walls.

Applications: Radio and television broadcasting, mobile communication, and wireless networks.

2. Microwaves

Working: Microwave signals are commonly transmitted using directional antennas. Terrestrial links usually require line of sight.

Advantages: High-capacity point-to-point communication and support for satellite links.

Applications: Satellite communication, cellular backhaul, and microwave relay systems.

3. Infrared

Working: Infrared transmitters send short-range infrared radiation to a receiver.

Advantages: Useful for short-range communication and generally does not pass through opaque walls.

Applications: TV remote controls, infrared sensors, and short-range device communication.

Comparative summary table

Feature	Twisted pair	Coaxial	Fiber optic	Radio waves	Microwaves	Infrared
Medium	Copper wires	Copper conductor and shield	Glass or plastic fiber	Air or space	Air or space	Air or space
Signal	Electrical	Electrical	Optical	Electromagnetic	Electromagnetic	Electromagnetic
Bandwidth	Moderate to high	High	Very high	Depends on frequency and system	Generally high	Depends on system
Noise immunity	Moderate	Good	Excellent	Subject to interference	Subject to interference	Low electrical interference; can be blocked by objects

Directionality	Guided	Guided	Guided	Often omnidirectional	Usually directional	Often directional or room-confined
Cost	Low	Moderate	Higher initial installation cost in many cases	Varies	Varies	Generally low for simple systems
Applications	Ethernet, telephone	Cable TV, broadband	Fiber internet, backbones	Broadcasting, mobile networks	Satellite, point-to-point links	Remote controls, sensors

Conclusion: Guided media provide a defined physical transmission path, while unguided media enable wireless communication. Fiber optic offers excellent bandwidth and noise immunity, twisted pair is economical, coaxial provides good shielding, and wireless media support mobility and flexible deployment.

Last-minute exam preparation tips

For each 7-mark answer, try to include:

1. A clear definition.
2. The main components or working steps.
3. Important technical points and examples.
4. Relevant formulas with variable meanings.
5. A comparison table wherever the question asks for differences.
6. A short concluding statement.

Since you do not need figures, focus on definitions, numbered points, formulas, examples, and comparison tables.